Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Legendre differential equations Continued

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Legendre differential equations Continued

1 Legendre differential equations Continued

Study focus: Legendre differential equations_Continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000186 · EV0000187

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will be solving problems on Legendre's linear differential equation and Cauchy's linear differential equation.

Key details

- In the previous session, we discussed about two different forms of linear differential equation with variable coefficients.
- One is legendary's linear differential equation, and the other one is.
- Cauchy's linear differential equation.
- We also learn method to solve Cauchy's and Legendary's linear differential equation.
- A quick recap of the concept studied in the previous session.
- What is this legendary linear differential equation?
- Our differential equation of the form k naught into a x plus b whole par n into nth order derivative of y with respect to x plus.
- K1 into a x plus b whole power n minus 1 into n minus 1 for the derivative of y with respect to x plus so on.
- Plus kn of y is equal to capital X where capital X is a function of the independent variable X, K0, K1, K2, K3, Kn are constants is legendary's linear differential equation of order n.
- See here, the degree of the coefficient is equal to the order of the derivative in all the terms.
- constant coefficients by the substitution z is equal to log a x plus b or e to the power z is equal to a x plus b or e to the power z minus b by a is equal to x.
- In the previous session, we derived that a x plus b into dy by dx is equal to a into d of y by capital.
- D stands for d by d dz.
- It is the differential operator with respect to z because.

- By using this substitution, we are changing the independent variable from X to Z.
- We also derived a x plus b the whole square into d2 y by dx2 is a square into d into d minus 1 of y, and a x plus b the whole cube into d3 y by dx3 is a cube into d into d minus 1 into d minus 2 of y.
- The differential equets transformed into LDE with constant coefficients, and the differential equation is solved by the methods which we learnt in the previous.
- We learned that is Cauchy's linear differential equation.
- We learned that Cauchy's linear differential equation is a special case of.
- Legendary C wherein a is equal to 1 and B is equal to 0.
- derivative of y with respect to x plus kn into y is equal to capital X where K0 K1 K2 K3 Kn are constant and K0 not equal to 0 and the right hand side function is a pure function of X Cauchy's LDE of all.
- The same method is used.
- With variable coefficient is reduced to differential equation with constant coefficients by using the substitution z is equal to log x or x is equal to e power z and x into dy by dx is equal to d of.
- X square where capital D stands for d by dz.
- X square into d2 y by d2 y by dx2 is d into d minus 1 of y.
- X cube d3 y by dx3 is d into d minus 1 into d minus 2 of y.
- Using these equations back in the differential equation, the equation gets transformed into LDE with constant coefficients, and the equation can be solved by the methods which we learned in the previous sessions.

Example or application from the lecture

- Now because in this session we are solving problems on Legendary's LDE and Cauchy's LDE, let us have.
- Using these equations back in the given legendary linear differential equation.
- Now let us apply the method which is learned to solve problems on and legend raised linear differential equation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 05:00–09:59

It is LDE with variable coefficients.

Key details

- And we learned only two different forms of LDE with variable coefficients.
- This is Cauchy's linear differential equation of order 3.
- equation by doing the substitution z is equal to log x r x is equal to e power z.
- We also derived x into dy by dx is equal to df y x square capital D stands for d by dz, x square into d2 y by dx2.
- D into d minus 1 of y x cube d3 y by dx is d into d minus 1 into d minus 2 of y.

- Here RHS also has to be converted into a function of z because we are changing the independent variable from x to z.
- X is e power z.
- What is 1 by x?
- It is e power minus z.
- A differential equation in which y is the dependent variable and z is the independent variable.
- Multiply these two terms 2 d square minus 2 d plus 2 of y is equal to 10 into e power z plus e power minus z.
- Simplifying it further, I get dq minus d square plus two of y is equal to 10 into e power z plus e power minus z.
- What is this differential equation?
- It is LD of order 3 with constant coefficient.
- Now, because RHS of the differential equation is not equal to 0, it is a non-homogeneous LD with constant coefficients and of order 3.
- How do I solve this differential equation?
- I know that the non-homogeneous LDE with constant coefficients has two components.
- One is the complementary function, and the second component is the particular integral.
- function and particular integral of this differential equation.
- Now next to find complementary function.
- I need to write the associated auxiliary equation.
- Is it cubic equation three roots now how do I find the roots see just by infect inspection I can find that m is equal to minus one is one root m is equal to minus one because it is minus one.
- And this is minus 1 minus 2 plus 2 equal to 0.
- Once you get one root, the other two roots are found by factorizing.
- The roots are m is equal to minus 1, comma, 1 plus r minus i.
- One is a real root and the other two roots are complex.
- One real root and two complex roots.
- The complementary function y c is equal to c 1 e to the power minus 1 into z.
- Ah, here it is not x because the independent variable in the differential equation is z.
- C1 E power minus Z plus.
- For this now I'm writing P par Z into C2 cos 1 Z plus C3 psi 1 Z, but.
- The complementary function has to be written in terms of x itself.
- 1 by x.
- What is e power plus z?
- It is x.
- Sign of log x.
- See here you should note that the final answer should be in terms of x itself.
- Next is to find the particular integral.
- It is 10 into e power z plus e power minus z divided by d q minus d square plus 2.

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EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

Core explanation · 10:00–14:59

It is 10 into e power z by d cube minus d square plus 2 plus 10 into e power minus z by d cube minus d square plus 2.

Key details

- See, I know that d is equal to minus 1 is a root of dq minus d square plus 2.
- I need to differentiate this, put 1x over here.
- We know that d is equal to 1 is not a root, so the denominator is definitely not equal to 0.
- 1 minus 1 plus 2, which is source 2.
- Now for the second term because the denominator is 0, what I need to do.
- So the first term is 10 e power z by 1 cube minus 1 square plus 2.
- Here I am writing z in 2.
- 10 e power minus z by dq.
- This dq minus d square plus 2 has to be differentiated with respect to d.
- It is 3d square minus 2d.
- See it is z into.
- Please note that it is not x into.
- Because the independent variable here is z, but not x.
- I'll do the substitution d is equal to minus 1.
- I get 10 e power z by 2 plus z into 10 e power minus z by 3.
- into d is equal to minus 1 minus 1 square minus 2 into minus 1.
- The denominator is not equal to 0's.
- Complete solution of the differential equation it is C F plus PI.
- It is it should not be in terms of Z.
- Solve 2x plus 3 the whole square, d2y by dx2 minus 2 into 2x plus 3, dy by dx minus 12y is equal to 6x.
- 2x plus 3 the whole square into second order derivative minus 2 into 2x plus 3 to the power 1 into dy by dx minus 12 into y is equal to 6x is the independent variable.
- See log to the base e.
- We also derived a 2x plus 3, that is a x plus b into dy by dx is e into d of y.
- I need the second one, 2.
- Where capital D stands for D by Dz.
- This has to be written where D is equal to D by C D Z.
- It is not D by DX.
- What is this?
- 4 into d into d minus 1 of y minus 2 into 2.
- The into d of y minus 12 of y is equal to 6 into r. x has to be written in terms of z using this equation.
- E power z minus 3 by 2.
- What is this differential equation?
- It is a linear differential equation with constant coefficients.
- To find complementary function, I need to write the associated auxiliary equation, which is m square minus 2m minus 3 is equal to 0.

Example or application from the lecture

- Let us consider the first return.

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EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

Core explanation · 15:00–19:59

To find the roots, the roots are 3 comma minus 1.

Key details

- It is in terms of z.
- But I need to reuse this equation 2x plus 3 is equal to e.
- e power z and write it in terms of x r x is equal to e power z minus three by two.
- How did I write this?
- The numerator 3 to the power z minus 9 is split it is split into two terms.
- Now, how do I simplify the first one?
- It's very simple.
- I just have to replace D by 1.
- Because it is e to the power a is a is 1, just put in the place of d you just put 1.
- Constant by f of d.
- 9 into e power 0z.
- divided by 4 into d square minus 2 d minus 3.
- Replace D by 0.
- D by 1 in the second term I have replaced D by 0.
- So this is because 1 is written as e to d by 0 and here because it is 0 into Z, so it is I have to just replace D by 0.
- But you need to write epar z in terms of s, which is 2x plus 3.
- into 2x plus 3 is the particular integral.
- Please note that when you are reporting the final solution, the complementary function and particular integral should be a function of x, which is the independent.
- Find the general solution.
- It is x into fourth order derivative of y with respect to x plus 10 into third order derivative of y with respect to x is equal to 0.
- By just looking at the differential equation, we say that it's a homogeneous differential equation.
- And but this differential equation doesn't resemble.
- I either Cauchy's or Legendary's linear differential equation, but this equation can be brought to Cauchy's linear differential equation.
- By multiplying the entire differential equation with x to the power 3 y.
- See it is x into 4th order derivative of y with respect to x.
- I want because the order of the derivative here is 4, I want x to the power 4 here.
- And it is 10 into x to the power 0 into third order derivative of y with respect to x.
- Because it is third order derivative, the coefficient of the.
- the degree of the coefficient should be 3.

- See multiplying the entire equation with x cube, it becomes x to the power 4 into d4 y by d.
- X4 plus 10 x cube into third order derivative of y with respect to x is equal to 0.
- Again by doing the substitution.
- x is equal to e to the power z, or z is equal to log x.
- X into d y by dx is equal to d of y where d is equal to this has to be mentioned where capital D is equal to d by dz.

Example or application from the lecture

- This is the complementary function of the given less generalized linear differential equation next is to find the particular integral.
- Variable of the given differential equation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

Core explanation · 20:00–24:59

X square into D2 Y by Dx2 is D into D minus 1 of Y.

Key details

- I need up to this four therivative.
- X cube into D3 Y by Dx3 is D into D minus 1 into D minus 2 of Y and X to the power 4 into 4th order derivative of y with respect to X is D into D.
- into d minus 1 into d minus 2 into d minus 3 of y.
- I get this term d into d minus 1 into d minus 2 into d minus 3 of y plus 10 into d into d minus 1 into d minus 2 of y is equal to 0.
- No need to multiply the linear factor.
- What are the common factors?
- fact is d into d minus 1 into d minus 2 can be taken as common fact.
- So I get d into d minus 1 into d minus 2 what is remaining the first term d minus 3 plus 10.
- Of y is equal to 0.
- Simplification would give d into d minus 1 into d minus 2 into d plus 7 of y is equal to 0.
- Now I need to this because this is a homogeneous LDE with constant coefficient and.
- I do I need not have to find the particular integral.
- I need to find only the.
- Complementary function and I know that the complementary function represents the complete solution of the homogeneous linear differential equation.
- The auxiliary equation is m into m minus 1 into m minus 2.
- To m plus 7 is equal to 0.
- The roots are 0, comma 1, comma 2, comma minus 7.
- So I have written I have not written y c I have written y because it is a homogeneous differential equation.
- C4 e to the power minus Z, but the final solution has to be in terms of X itself.
- This is how we find the general solution of Cauchy's and Legendary's linear differential equation.

- So, we learn method to find the general solution of Cauchy's and Legendary's linear differential equation.

Example or application from the lecture

- Let us take a common factor from the two terms.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000202 · EV0000203 · EV0000204 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Legendre differential equations_Continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equations_Continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Legendre differential equations_Continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Legendre differential equations_Continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/011_Legendre_differential_equations_Continued__e286e5075a9d86f4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00046	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00047	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00048	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00049	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

C Revision and retrieval practice